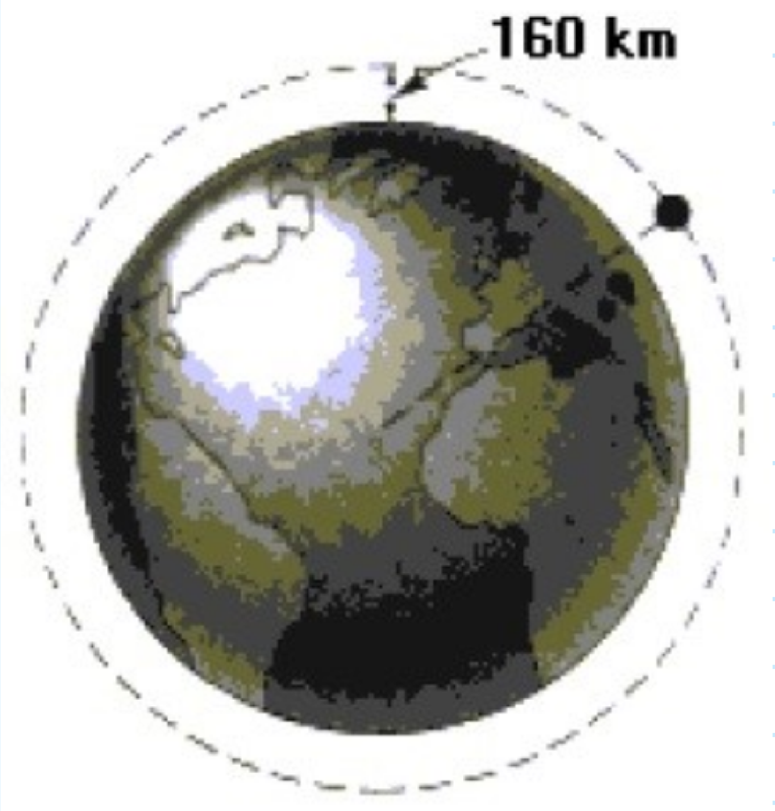
Ligjerata 7e

Perioda e rrotullimit te satelitit



$$\frac{T^2}{r^3} = \frac{4\pi^2}{\gamma M} \qquad \gamma M = g^0 R^2$$

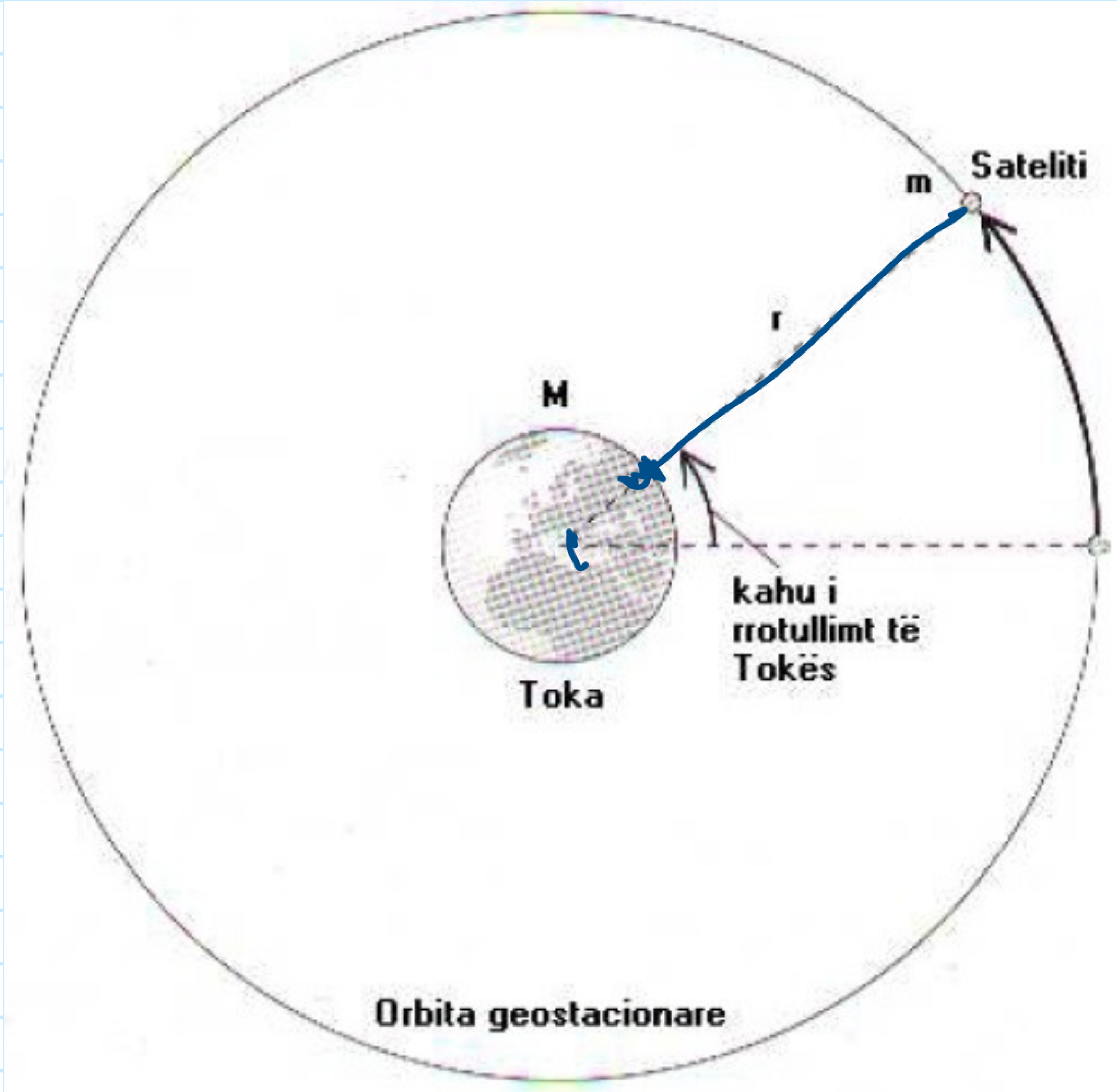
$$\frac{T^2}{r^3} = \frac{4\pi^2}{g^0 R^2} \quad / \cdot r^3$$

$$T^2 = \frac{4\pi^2}{g^0 R^2} \cdot r^3 \quad / \sqrt{}$$

$$T = \sqrt{\left(\frac{2\pi}{R}\right)^2 \cdot \frac{r^3}{g^0}} = \underline{\underline{r\sqrt{\left(\frac{2\pi}{R}\right)^2 \frac{r}{g^0}}}}$$

$$\sqrt{r^3} = \sqrt{r^2 \cdot r} = \sqrt{r^2} \cdot \sqrt{r} = \underline{\underline{r \cdot \sqrt{r}}}$$

Satelitët gjeostacionarë



$$\frac{T^2}{r^3} = \frac{4\pi^2}{\gamma M} \qquad \gamma M = g^0 \cdot R^2$$

$$r^3 = \frac{T^2 \cdot g^0 R^2}{4\pi^2} \quad / \sqrt[3]{}$$

$$\sqrt[3]{r^3} = \sqrt[3]{\frac{T^2 g^0 R^2}{4\pi^2}}$$

$$r = \sqrt[3]{\frac{T^2 g^0 R^2}{4\pi^2}}$$

$$r = \sqrt[3]{\left(\frac{T \cdot R}{2\pi}\right)^2 \cdot g^0}$$

$$\boxed{r = R + h}$$

$$R + h = \sqrt[3]{\left(\frac{TR}{2\pi}\right)^2 \cdot g^0}$$

$$\boxed{h = \sqrt[3]{\left(\frac{TR}{2\pi}\right)^2 \cdot g^0} - R}$$

$$T = 24h = 86400\,s$$

$$R = 6{,}378 \cdot 10^6\,m$$

$$g = 9{,}81\,\frac{m}{s^2}$$

$$h = 35900\,km$$

$$\boxed{h \approx 36000\,km}$$